YOUSEF AHMED

Junior AI Engineer

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PROFESSIONAL SUMMARY

Mechatronics Engineering graduate (GPA 3.62) with hands-on experience building AI systems across computer vision, deep learning, and predictive modeling. Skilled in Python, TensorFlow, and YOLO-based object detection, with practical knowledge of MLOps workflows. Published researcher with a proven ability to take AI projects from concept to deployment — including an autonomous agricultural robot achieving 96% accuracy in the field.

EXPERIENCE

Machine Learning Engineer (Virtual Training) *Jun 2024 – Aug 2024*
PioPetro • Ohio, United States (Remote)

- Completed a virtual training program on ML applications in the petroleum industry, covering Petroleum Exploration, Artificial Lift Technology, Well Intervention, and Well Stimulation.

AI/ML Intern *Jul 2023 – Sep 2023*
Information Technology Institute (ITI) • Al Minufiyah, Egypt

- Applied Python for data analysis, preprocessing, and ML model implementation across supervised and unsupervised tasks.
- Deepened knowledge of linear algebra, probability, and statistics as foundations for model development.

AI/ML Intern *Aug 2023*
Digital HUB (D-HUB) • Cairo, Egypt

- Built a breast cancer detection model achieving 95% accuracy, earning 3rd place in a competitive AI challenge co-hosted with the Electronics Research Institute.

PROJECTS

Agri-Bot — AI-Powered Crop Disease Detection Robot (Graduation Project) *Nov 2023 – Jun 2024*

- Designed and deployed an autonomous agricultural robot using YOLOv8 and MobileNetV2 on NVIDIA Jetson Nano, achieving 96% disease detection accuracy at 20 FPS.
- The system recommends targeted crop treatments, demonstrating real-world impact for agricultural sustainability in Egypt.

Ball Tracking Mobile Robot *May 2023*

- Built a computer vision-based robot achieving 95% real-time object detection accuracy with sub-5cm tracking error using autonomous navigation algorithms.

Steganography Image Detection *Feb 2023*

- Developed a deep learning system to detect hidden data embedded in digital images, achieving 90% classification accuracy.

Automatic Color Sorting Machine *Jun 2022*

- Engineered an AI-driven sorting machine with 98% color classification accuracy, improving sorting throughput by 25%.

EDUCATION

B.Sc. Mechatronics Systems Engineering *2020 – 2024*
MSA University (supervised by University of Greenwich) • Giza, Egypt • GPA: 3.62 / 4.0

SKILLS

Programming: Python | C++ | MATLAB
ML / DL: Supervised & Unsupervised Learning | NN | Transfer Learning | CNNs | YOLOs
Libraries: Pandas | NumPy | Matplotlib | Seaborn | Scikit-learn | FastAPI | TensorFlow
MLOps: Git | GitHub | Docker | Kubernetes
Soft Skills: Problem-solving |Time management | Adaptability | Researcher

PUBLICATIONS

Agri-Bot: AI-Powered Crop Disease Detection and Medication Recommendation Robot
Published at the 8th International Undergraduate Research Conference (IUGRC 8) • 2024

CERTIFICATIONS

- Machine Learning Specialization — Coursera (2023)
- Supervised Machine Learning: Regression and Classification — Coursera (2023)
- Python 101 for Data Science — IBM Cognitive Class (2022)